



Autoignition study of *iso*-cetane/tetralin blends at low temperature

Mengyuan Wang^a, Goutham Kukkadapu^{b,*}, Ruozhou Fang^{a,*}, William J. Pitz^b, Chih-Jen Sung^a

^a Department of Mechanical Engineering, University of Connecticut, Storrs 06269, CT, USA

^b Materials Science Division, Lawrence Livermore National Laboratory, Livermore 94551, CA, USA

ARTICLE INFO

Article history:

Received 14 December 2020

Revised 15 February 2021

Accepted 15 February 2021

Available online 2 March 2021

Keywords:

iso-cetane

Tetralin

Rapid compression machine

Autoignition

Chemical kinetic model

Low temperature combustion

ABSTRACT

iso-Cetane and tetralin are the two fuel components commonly considered in literature diesel surrogates, and are hydrocarbon classes representative of *iso*-alkanes and naphthoaromatics, respectively. Since both surrogate components are involved in the key reactions controlling the autoignition reactivity of diesel surrogates, autoignition studies of *iso*-cetane, tetralin, and their blending behavior at low temperatures can be helpful in developing/refining chemical kinetic models of diesel surrogates, in order to better predict diesel ignition response under low temperature combustion conditions. In this investigation, autoignition experiments of *iso*-cetane, tetralin, and their binary mixtures at varying blending ratios have been conducted in a rapid compression machine (RCM) in the temperature/pressure range of 630–930 K and 10–20 bar. For the neat fuel components, the newly-acquired RCM data have been compared with the literature experimental results at some overlapping conditions. In addition, a chemical kinetic model describing the oxidation of *iso*-cetane and tetralin compiled/updated from the diesel surrogate model developed by Lawrence Livermore National Laboratory is used to simulate the experimental datasets. It is shown that this model exhibits good agreement with the experimental results of tetralin and also predicts the total ignition delay times of *iso*-cetane at lower temperatures reasonably well. However, the current model fails to fully capture the increase of total ignition delay time with increasing amount of tetralin in the binary blends observed in the present RCM experiments. Therefore, chemical kinetic analyses of the current model are conducted to help identify possible reasons leading to the discrepancies between simulations and experiments. Furthermore, the chemical kinetic interactions between *iso*-cetane and tetralin are explored by including additional cross-reactions among *iso*-cetane, tetralin, and their fuel fragments in the current model to assess their effects on model predictions. Based on the chemical kinetic analysis results, future direction for model refinements is discussed.

© 2021 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

1. Introduction

As advanced diesel engines operated at low temperature combustion conditions are currently being developed to reduce the emissions of nitrogen oxides and particulate matter, as well as to burn the fuel more efficiently [1], the fundamental understanding of the autoignition characteristics of diesel fuels at low-to-intermediate temperatures becomes important. Considering diesel fuels contain even thousands of species, the development of diesel surrogates, which are simpler fuel mixtures containing approximately 10 or fewer pure “palette” compounds but can capture the essential chemical/physical properties and performance character-

istics of their target diesel fuels to sufficient accuracy [2], and the development of their chemical kinetic models are vital in predicting autoignition and combustion of diesel fuels and reducing the design cycle of future diesel engines. As an example, Mueller et al. [3] provided a methodology for formulating diesel surrogate fuels with improved fidelity in matching the molecular structures, ignition quality, and distillation characteristics of two target diesel fuels, namely a 2007 emissions certification diesel fuel and a Coordinating Research Council (CRC) Fuels for Advanced Combustion Engines (FACE) diesel fuel. In a follow-up study by the CRC, Mueller et al. [2] further formulated, blended, and characterized a set of four diesel surrogate fuels, including 4-component V0a, 5-component V0b, 8-component V1, and 9-component V2, which were designed to emulate a certified ultra-low-sulfur research-grade diesel supplied by Chevron Phillips Chemical. An autoignition study of the four CRC diesel surrogates has been conducted

* Corresponding authors.

E-mail addresses: kukkadapu1@llnl.gov (G. Kukkadapu), ruozhou.fang@uconn.edu (R. Fang).

by Wang et al. [4] at engine relevant conditions with lean to rich fuel/oxidizer mixtures in a rapid compression machine (RCM), and the experimental data of the CRC diesel surrogates were also used to validate a detailed diesel surrogate chemical kinetic model [5] developed by the Lawrence Livermore National Laboratory (LLNL). By comparing the simulated results with the experimental data of the CRC diesel surrogates, Wang et al. [4] showed that while the performance of this LLNL diesel surrogate model is reasonable in predicting the overall ignition characteristics, further refinements are still needed for better agreement, especially for total ignition delay times in the low temperature regime (LTR) and first-stage ignition delay times in general. Since this LLNL diesel surrogate model was assembled by merging the individual models of the constituent hydrocarbons, it may not be able to fully capture the chemical kinetic interactions when the surrogate constituents are blended together, which could be a possible source of discrepancy and hence merits further investigation.

In this study, the binary blends of *iso*-cetane (2,2,4,4,6,8,8-heptamethylnonane) and tetralin (1,2,3,4-tetrahydronaphthalene) are selected for the experimental investigation of their ignition behaviors at varying blending ratios. *iso*-Cetane and tetralin respectively represent the hydrocarbon classes of *iso*-alkanes and naphthoaromatics, which are among the major compound classes in the CRC diesel surrogates. From the sensitivity analysis results shown in Wang et al. [4], it was concluded that both *iso*-cetane and tetralin are the fuel components involved in main reactions governing the reactivity of the CRC diesel surrogates. In addition, the reaction pathway analyses of fuel components before and during the first-stage ignition event conducted in [4] showed that *iso*-cetane and tetralin have been consumed almost simultaneously. As a result, the blending behavior between those two surrogate components are the emphasis of this study. Although the chemical kinetics for *iso*-cetane has been studied to some detail [6–12], there are not many studies focusing on autoignition of *iso*-cetane at low-to-intermediate temperatures [11,12]. Similarly, among the literature studies for tetralin [13–17], the studies investigating its autoignition chemistry at low-to-intermediate temperatures are meager [16]. Wilk [18] studied the oxidation of *n*-dodecane/tetralin mixtures in a constant volume vessel at 60–240 Torr and 520–620 K, and found that the addition of tetralin to *n*-dodecane increases the induction period before the appearance of cool flames. Moreover, this tetralin blending effect was noted in [18] to be nonlinear with the 5% addition of tetralin having the largest effect and further addition of tetralin having proportionally smaller effect. To the best of our knowledge, there is no autoignition study on the binary blends of *iso*-cetane and tetralin, and hence it is worth studying whether a similar nonlinear effect on autoignition also exists at conditions of elevated pressure and low-to-intermediate temperature when adding tetralin to *iso*-cetane.

Recognizing the need for more experimental studies on these two neat fuel components and their blends, autoignition experiments using an RCM are first conducted for neat components of *iso*-cetane and tetralin. The newly-acquired experimental results are then compared with the literature data under overlapping test conditions, followed by experimental autoignition characteristics investigation of *iso*-cetane/tetralin binary blends with varying blending ratios. Subsequently, these new experimental datasets are compared with the simulated results using a chemical kinetic model for *iso*-cetane/tetralin blends that has been taken from and refined based on the LLNL diesel surrogate model of Kukkadapu et al. [5]. Furthermore, chemical kinetic analyses of binary blends are conducted herein for better understanding of the dominant reactions controlling the reactivity and possible reasons accounting for discrepancies between experiments and simulations. As mentioned earlier, the LLNL diesel surrogate model [5] was assembled from the chemical kinetic models of surrogate constituents without

considering the cross-reactions among fuel components. Therefore, different classes of cross-reactions among *iso*-cetane, tetralin, and their fragments are surveyed and compiled. Furthermore, these cross-reactions are added to the current *iso*-cetane/tetralin model to examine the effect of cross-reactions on simulated autoignition results. The newly-acquired RCM datasets, the comparison of experimental and simulated results, and the chemical kinetic analyses conducted in this study shall not only provide insights into the blending behaviors of *iso*-cetane and tetralin, but also shed light on the refinements and development of comprehensive diesel surrogate models.

2. Experimental specifications

An RCM is used in this study to compress the desired gas mixture to elevated pressures and temperatures using a fast moving piston which is brought to rest towards the end of compression (EOC). Post compression, the piston is held stationary thus providing a constant volume reactor. This RCM is designed to reduce both bulk fluid motion and thermal inhomogeneity, thereby offering an ideal platform for studying chemical kinetics and ignition characteristics of fuel/oxidizer mixtures. The present RCM consists of a pneumatic cylinder, a hydraulic cylinder, and a reaction chamber, in which the reactant gas mixture is compressed by a creviced piston arrangement during each experimental run. First, the creviced piston in the reaction chamber is retracted from the top dead center to the bottom dead center. It is held in place by pressurized oil in the hydraulic cylinder. When the solenoid valve is triggered, the pressure in the hydraulic chamber is released and the creviced piston moves forward, driven by the high-pressure air tank and the pneumatic piston inside the pneumatic cylinder. The creviced piston is then decelerated and stopped by a stopping ring and groove mechanism in the hydraulic cylinder. The compression stroke of the current RCM typically takes a time in the range of 25–50 ms. A thermal-shock resistant, dynamic pressure transducer (Kistler 6125C) is equipped on the reaction chamber for dynamic pressure measurements during experiment, along with a charge amplifier (Kistler 5010B). The compression ratio of this RCM can be varied by changing the length of stroke or/and the clearance volume. Split shims are used to vary the clearance between the hydraulic cylinder and the reaction chamber, while spacers are employed to adjust the stroke length. More details about the RCM design, operation, and characterization can be found in [19,20].

Each of the *iso*-cetane/air, tetralin/air, and binary fuel blend/air mixtures is prepared in a stainless steel mixing tank. A magnetic stirrer is placed at the center of the tank bottom to provide a mixing mechanism. A rupture disc is used as a safety measure against accidental overpressure within the tank. For the current study, each mixture is prepared at room temperature, with liquid fuel being massed in a glass syringe to a precision of 0.01 g prior to injection into the mixing tank through a septum, followed by oxygen and nitrogen filled into the tank on a barometric measure in the molar proportion of $O_2:N_2=1:3.76$. The partial pressures of gaseous O_2 and N_2 filled in the mixing tank are measured by a static pressure transducer (Omega, MMA100V10T2D0T4A6, $\pm 0.05\%$ accuracy) located at the manifold of mixing tank. Recognizing that the gas heating effect can occur during the tank filling process as the line enthalpy is converted to tank internal energy, the tank temperature is kept within $\pm 1.5^\circ C$ of the room temperature when filling each gas to the target pressure. High purity of fuels and gases are used for this study, including *iso*-cetane (98% purity) and tetralin (99% purity) from Sigma-Aldrich, and O_2 ($> 99.99\%$ purity) and N_2 ($> 99.99\%$ purity) from Airgas. The mixing tank, the manifold, and the reaction chamber, all equipped with heating tapes, are heated to the desired preheat temperature of 420 K for 4 h before starting the experiments to ensure complete vaporization of

Table 1
Test conditions for iso-cetane/tetralin binary blends.

ϕ	X_{IC16} (%)	X_{tetralin} (%)	X_{O_2} (%)	X_{N_2} (%)	Molar% of tetralin in the fuel blend	P_C (bar)
0.5	0.43	0	20.92	78.65	0	20
2.0	1.69	0	20.65	77.66	0	10
1.0	0.85	0	20.83	78.32	0	15, 20
1.0	0.82	0.04	20.83	78.31	5	15
1.0	0.78	0.09	20.83	78.30	10	15, 20
1.0	0.71	0.18	20.82	78.29	20	15, 20
1.0	0.64	0.27	20.82	78.27	30	20
1.0	0.48	0.48	20.81	78.23	50	15, 20
1.0	0	1.12	20.77	78.11	100	15, 20

liquid fuel(s) and a homogeneous fuel/oxidizer mixture. It is also noted that when preparing the reactant mixture, the partial pressure of each fuel component, as well as that of binary fuel blend at each blending ratio, are kept less than half of the respective saturated vapor pressure at 420 K. In order to monitor the wall temperature uniformity of the reaction chamber, three thermocouples are used and evenly distributed along the outside wall of the reaction chamber. For the set preheat temperature of 420 K in the current study, the temperature readings of the three thermocouples are typically within 413–423 K.

It is further noted that the uncertainty of mixture preparation includes the uncertainties of liquid fuel mass, partial pressures of gaseous components, mixing tank volume, and ambient temperature, with their standard deviations being 0.03 g, 1.3 Torr, 0.01 L, and 1.27 °C, respectively, as discussed in Weber et al. [21]. Following the detailed uncertainty analysis method of [21], the errors in mole fractions for iso-cetane and tetralin are estimated to be 5–10% for most of the mixture conditions listed in Table 1, while those for gaseous O_2 and N_2 are typically in the range of 0.1–0.9%.

The compressed temperature (T_C), namely the gas temperature at the EOC, is used as the reference temperature in reporting the present RCM data, and is deduced from the measured pressure trace by using the assumption of “adiabatic core hypothesis” [22]. Under this hypothesis, it is assumed that the heat transfer from the core volume of the reaction chamber only occurs in a thin boundary layer near the wall, and the central core region remains adiabatic by modeling the heat transfer effect as an effective reduction in the compression ratio [23,24]. Thus, T_C can be calculated by $\ln(\frac{P_C}{P_0}) = \int_{T_0}^{T_C} \frac{\gamma}{(\gamma-1)T} dT$, where P_C is the measured pressure at the EOC, P_0 is the initial pressure, T_0 is the initial temperature, and γ is the temperature-dependent specific heat ratio. Following the estimation of Weber et al. [21], the uncertainty of the deduced compressed temperature is estimated to be within $\pm 0.7\%$ of the mean calculated temperature for the instruments installed on the present RCM, by considering the uncertainties of the initial pressure and temperature, the compressed pressure, the initial mixture composition, and the fits for the specific heats of the mixture components. For clarity, the error bars representing T_C uncertainty are not shown in the figures.

The test conditions investigated in the current study are outlined in Table 1, including equivalence ratio (ϕ), molar percentage of each mixture constituent (X_i), P_C , and the molar percentage of tetralin in the binary fuel blend. The RCM experiments for neat components of iso-cetane and tetralin have been designed to compare with the published data of [11,12,16] and also aim at varying pressures and covering a wide range of temperatures. In addition, the molar percentage of tetralin (%tetralin) in the stoichiometric binary fuel blend is systematically varied to investigate the response of the blend reactivity to the blending ratio. It should be mentioned that we focus less on the blends with %tetralin more than 50% because the total ignition delay times of these conditions

become too long (>200 ms) at low temperatures. We choose not to report data points that have total ignition delay times much longer than 200 ms because the residual vortex (if any) could eventually lead to temperature inhomogeneity in the core region of the reaction chamber after long experimental duration [25]. It is also noted from Table 1 that the molar percentages of O_2 and N_2 in the test mixtures, X_{O_2} and X_{N_2} , remain largely unchanged for the fuel/air mixture conditions investigated herein.

Figure 1(a) shows a representative pressure trace of the 10/90 tetralin/iso-cetane blend (10% tetralin and 90% iso-cetane by mole) in air at $\phi=1.0$, $P_C=20$ bar, and $T_C=666$ K, demonstrating the definitions of the first-stage ignition delay time (τ_1) and the total ignition delay time (τ). The EOC point is regarded as the starting point of ignition delay time, which is set as $t = 0$ ms. In Fig. 1(a), τ_1 and τ are both identified based on the respective local maximum of the time derivative of the pressure trace. The nonreactive pressure trace of the same experimental condition is also shown in Fig. 1(a) as a reference. For each reactive set of P_C and T_C , the corresponding nonreactive pressure trace is measured by replacing O_2 with N_2 to characterize the heat transfer effect on the autoignition process, as well as to verify that no heat release has occurred during the compression stroke. For each experimental condition, four to five consecutive runs are taken to ensure data repeatability, which can be seen in Fig. 1(b) for stoichiometric 10/90 tetralin/iso-cetane blend at $P_C=15$ bar and $T_C=851$ K. The typical scatter is found to be less than 15% of the reported ignition delay time (τ_1 or τ), and the scatter is defined as the ignition delay time range ($\pm\%$) of the consecutive runs based on the representative value. After taking the major uncertainties related to ignition delay time measurements into consideration, such as the uncertainties in reactant mixture preparation, compressed temperature (T_C) determination, and pressure measurement [24], the current ignition delay time data of all four to five consecutive runs for each test condition still fall into the 15% scatter range.

3. Computational specifications

The current detailed chemical kinetic model for iso-cetane/tetralin blends is taken from the diesel surrogate model of Kukkadapu et al. [5] by combining the sub-models of the two neat fuel components therein. It has to be pointed out that the sub-models of tetralin and iso-cetane used in the LLNL diesel surrogate model [5] are based on the studies of Issayev et al. [26] and Ritcher et al. [27], respectively. Some modifications are also implemented for the present autoignition study, namely removing the reactions related to n -alkanes larger than C_{10} and the duplicate reactions in the aromatics sub-model. In addition, according to the results and discussions in [4,5], the reaction rates of H-abstraction of iso-cetane by OH radicals are updated following the studies of [28,29]. This chemical kinetic model for blends of iso-cetane and tetralin, consisting of 3137 species and 12,506 reactions, is included in Supplementary Material.

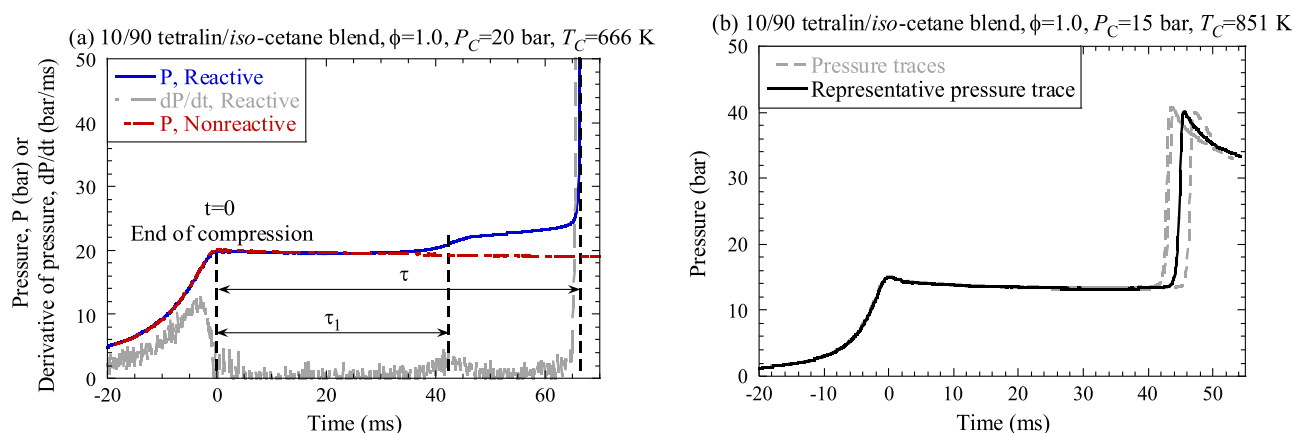


Fig. 1. (a) Plot showing the definitions of first-stage and total ignition delay times and the corresponding nonreactive pressure trace. (b) Plot showing the experimental repeatability of the current data.

RCM simulations have been performed using the current binary blend model in Chemkin-Pro [30]. As described in [24], the heat transfer effects during the compression stroke and after reaching the EOC are accounted for by using volume traces derived from nonreactive experiments.

4. Results and discussion

4.1. Comparison of pure fuel experimental data to modeling results and literature data

Figure 2 compares the current experimental autoignition results of two neat components with the literature RCM data under various pressures and equivalence ratios. In particular, Figs. 2(a)–2(d) compare the total and first-stage ignition delay times of iso-cetane at $P_C=10$ –20 bar and $\phi = 0.5$ –2.0 for the current RCM results and the literature RCM data taken from Yu et al. [12]. Additionally, the RCM iso-cetane data of Kukkadapu and Sung [11] are also included in Fig. 2(a) for comparison. It is noted that the experimental data uncertainties reported in [11] and [12] are 15% and 10%, respectively. As discussed earlier, by comparing with the corresponding nonreactive pressure traces, the RCM data exhibiting exothermicity during the compression stroke can be identified. For the current iso-cetane results shown in Figs. 2(b) and 2(d), those data points with heat release during the compression stroke are excluded.

It can be observed from Fig. 2(a) that the current iso-cetane/air data in the intermediate temperature regime (ITR) is comparable with those of Kukkadapu and Sung [11] at $P_C=15$ bar and $\phi = 1.0$. However, the total ignition delay times obtained herein are consistently faster than the experimental data of Yu et al. [12] at fuel lean to stoichiometric test conditions shown in Figs. 2(a)–2(c), especially in the negative temperature coefficient (NTC) regime and ITR. For the test conditions of $P_C=10$ bar and $\phi = 2.0$ in Fig. 2(d), the current total ignition delay times are longer than those of Yu et al. [12] over the entire temperature range investigated. One possible reason causing the discrepancies between the current experimental data and those of Yu et al. [12] is the difference in initial temperature used in RCM experiments. The initial temperature set in this study is $T_0 = 420$ K for all test conditions, as used in Kukkadapu and Sung [11], while in Yu et al. [12] $T_0 = 393$ K for the test conditions in Figs. 2(a)–2(c) and $T_0=423$ K for the test conditions in Fig. 2(d). For the first-stage ignition delay times, on the other hand, the current RCM results are comparable with the literature data for all test conditions in Figs. 2(a)–2(d).

Figures 2(e)–2(f) further compare the total ignition delay times of stoichiometric tetralin/air at $P_C=15$ bar and 20 bar for the cur-

rent experimental results, the RCM data taken from Kukkadapu et al. [16], and the shock tube data reported by Wang et al. [14] and Raza et al. [17]. It is noted that the reported uncertainties of total ignition delay times are 10% in [16] and 20% in [14,17]. In addition, the shock tube data have been scaled to the current experimental pressure conditions using the pressure dependence discussed in [14,17]. It is seen that the current experimental data are comparable with the literature results of Kukkadapu et al. [16] within their uncertainties. The current RCM data also compare well with the shock tube data of [14], while the shock tube data of [17] show consistent shorter total ignition delay results than those obtained from RCM, especially at temperatures below 1000 K. This discrepancy at low temperatures may be attributed to the steady increase of pressure prior to ignition in shock tube experiments that was not specified in [17].

The above-mentioned experimental ignition delay times have then been compared with the simulated results using the current binary blend model, as shown in Fig. 2. It can be observed from the comparison between experiments and simulations of iso-cetane that the discrepancies are the largest in the ITR, with over-prediction up to a factor of 4. These discrepancies may be due to the C_0 – C_4 base mechanism from AramcoMech 2.0 [31] that needs further refinements as has also been discussed in [4]. The simulated results agree better with the experimental data in the LTR and NTC regime, with discrepancies being up to a factor of 2. From the comparison of experimental and simulated first-stage ignition delay times, it indicates that larger discrepancies in first-stage ignition delay times typically result in larger discrepancies in total ignition delay times. In addition, the simulation results of both total and first-stage ignition delay times using the conditions of the current study are consistently faster than those obtained using the conditions of Yu et al. [12]. This trend for the simulation results is consistent with the experimental results for the total ignition delay times in Figs. 2(a)–2(c), but not for the experimental total ignition delay times in Fig. 2(d) and first-stage ignition delay times. It is also noted from Fig. 2(d) that the experimental total ignition delay time comparison between the current study and Yu et al. [12] shows the opposite trend at $P_C=10$ bar and $\phi = 2.0$, while the experimental first-stage ignition delay times of the current study and Yu et al. [12] are comparable within their uncertainties at all test conditions investigated.

To further assess the current iso-cetane sub-model at high temperatures, Fig. 3 compares simulated and experimental laminar flame speeds of iso-cetane/air mixtures at atmospheric pressure and unburned mixture temperature of 443 K. The simulated laminar flame speeds are computed using the ZeroRK solver [32], while

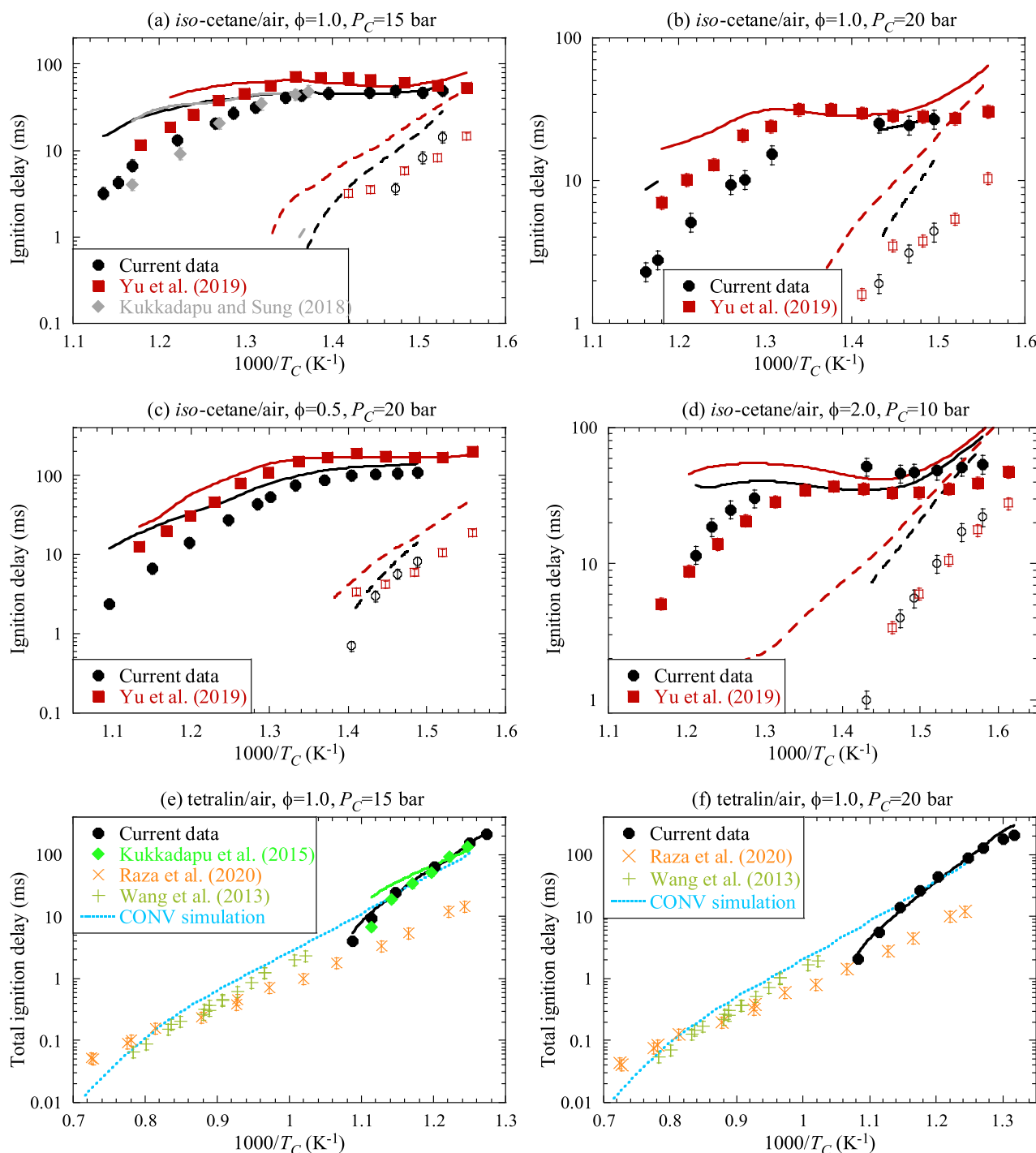


Fig. 2. Comparison of experimental ignition delay times of *iso*-cetane/air (a–d) and tetralin/air (e,f) at various pressures and equivalence ratios from the current study and literature [11,12,14,16,17], and their simulation results using the current binary blend model. Filled symbols: experimental total ignition delay times from RCM experiments. Open symbols: experimental first-stage ignition delay times from RCM experiments. Cross symbols: experimental total ignition delay times from shock tube experiments. Solid lines: RCM simulated total ignition delay times. Dashed lines: RCM simulated first-stage ignition delay times. Dotted lines: CONV simulated total ignition delay times.

the experimental data are taken from Li et al. [33]. It is seen from Fig. 3 that the simulated laminar flame speeds are consistently lower than the experimental data for all equivalence ratios investigated, with discrepancies up to 6 cm/s (~16%) compared to the reported experiment uncertainty of $\pm 3\%$ or ± 1.2 cm/s at $\phi = 1.4$ [33]. As has been stated in [33] that laminar flame speed is largely sensitive to C₀–C₄ kinetics for *iso*-cetane/air mixtures, refinements of Aramco2.0 [31] may improve the laminar flame speed prediction. This improvement may also result in better prediction of total igni-

tion delay times of *iso*-cetane at higher temperatures, as discussed earlier.

For the simulated total ignition delay times of tetralin, it can be seen from Fig. 2(e) and 2(f) that the predicted values using the current binary blend model agree reasonably well with the experimental data of both the current study and Kukkadapu et al. [16] at temperatures lower than 925 K. Adiabatic constant-volume (CONV) simulation results have also been shown in Figs. 2(e) and 2(f) to compare with the literature shock tube data. It is found that the

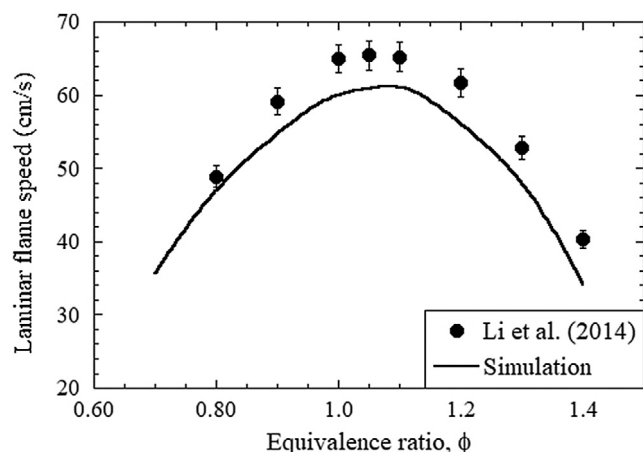


Fig. 3. Comparison of experimental [33] and simulated laminar flame speeds of iso-cetane/air mixtures at unburned mixture temperature of 443 K and atmospheric pressure.

CONV results agree better with the experimental data of [14] than those of [17].

4.2. Comparison of binary blend experimental data to modeling results

The experimental total and first-stage ignition delay times with varying iso-cetane/tetralin blending ratios at $P_C = 15$ and 20 bar

and $\phi = 1.0$ are shown in Fig. 4, with the test conditions being listed in

Table 1. It can be observed that both total and first-stage ignition delay times increase with increasing tetralin in the binary blend at lower temperatures, and the NTC characteristics can be clearly seen in the total ignition delay time results for the blends containing less than 30% tetralin. For the binary blend with 50% tetralin, only data points with T_C higher than 785 K at $P_C = 15$ bar and data points with T_C higher than 735 K at $P_C = 20$ bar have been recorded to avoid reporting total ignition delay times being much longer than 200 ms as mentioned previously. For $P_C = 15$ bar shown in Fig. 4(a), a noticeable increase of experimental total ignition delay times can be seen as tetralin increases from 0% to 5%, while further increase of tetralin in the binary fuel blend has smaller effect on total ignition delay times especially at higher temperatures, with the data points overlapping in the ITR. Similar total ignition delay time trend can also be seen in Fig. 4(b) for $P_C = 20$ bar as tetralin increases noticeably from 0% to 10% and continues increasing from 10% to 100%. This reactivity trend is consistent with the blending effect of adding tetralin to *n*-dodecane previously observed in [18].

RCM simulation results of iso-cetane/tetralin blends using the current model have also been shown in Fig. 4 and compared with experimental data at various blending ratios. It can be noticed that certain simulated results of iso-cetane at $P_C = 20$ bar have not been shown in Fig. 4(b), which is because exothermicity during the compression stroke is noted in the simulated pressure traces by comparing with the corresponding nonreactive experimental pressure

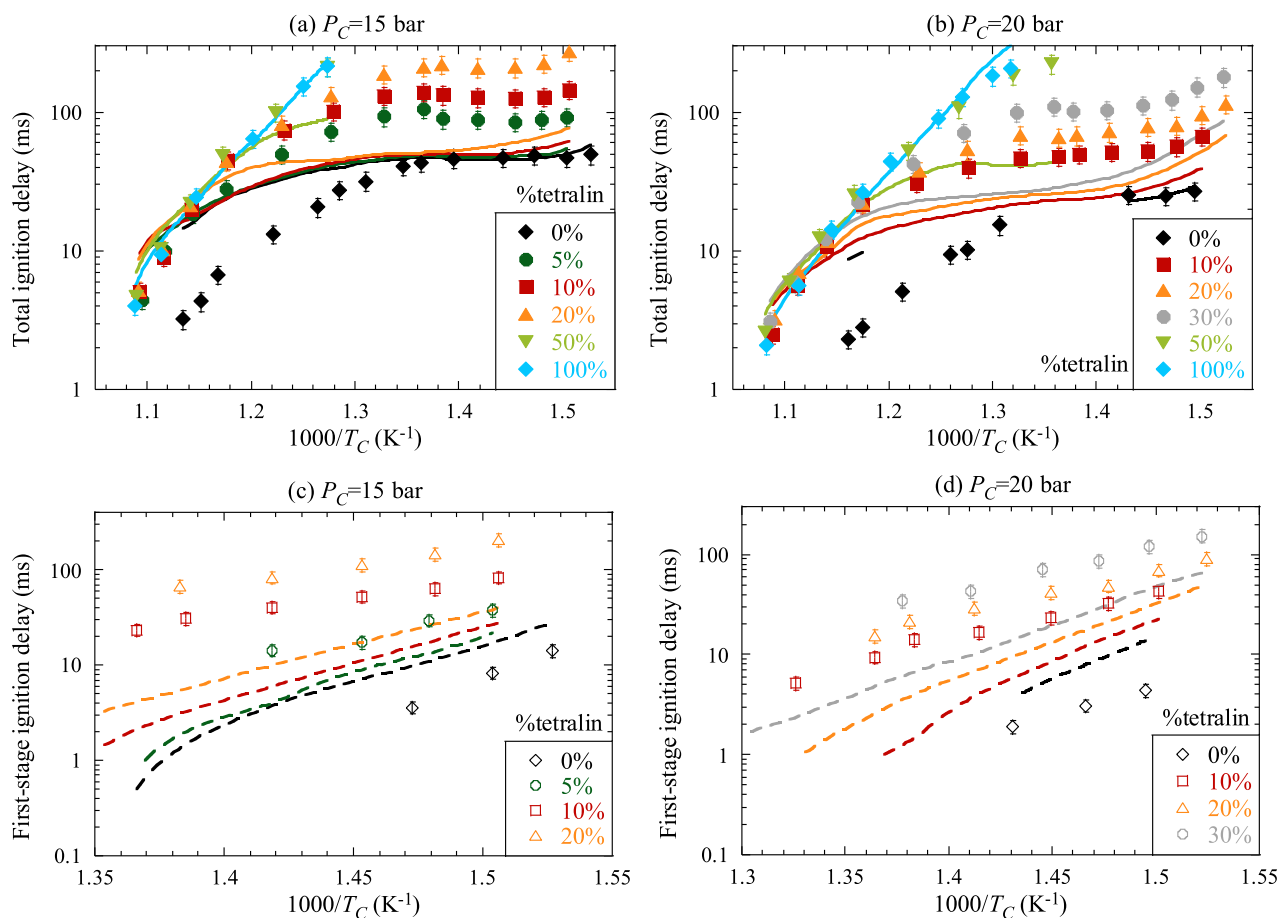


Fig. 4. Experimental total and first-stage ignition delay times of stoichiometric iso-cetane/tetralin blends at various blending ratios and $P_C = 15$ and 20 bar, as well as the comparison with simulated ignition delay times using the current binary blend model. Solid lines: simulated total ignition delay times. Broken lines: simulated first-stage ignition delay times.

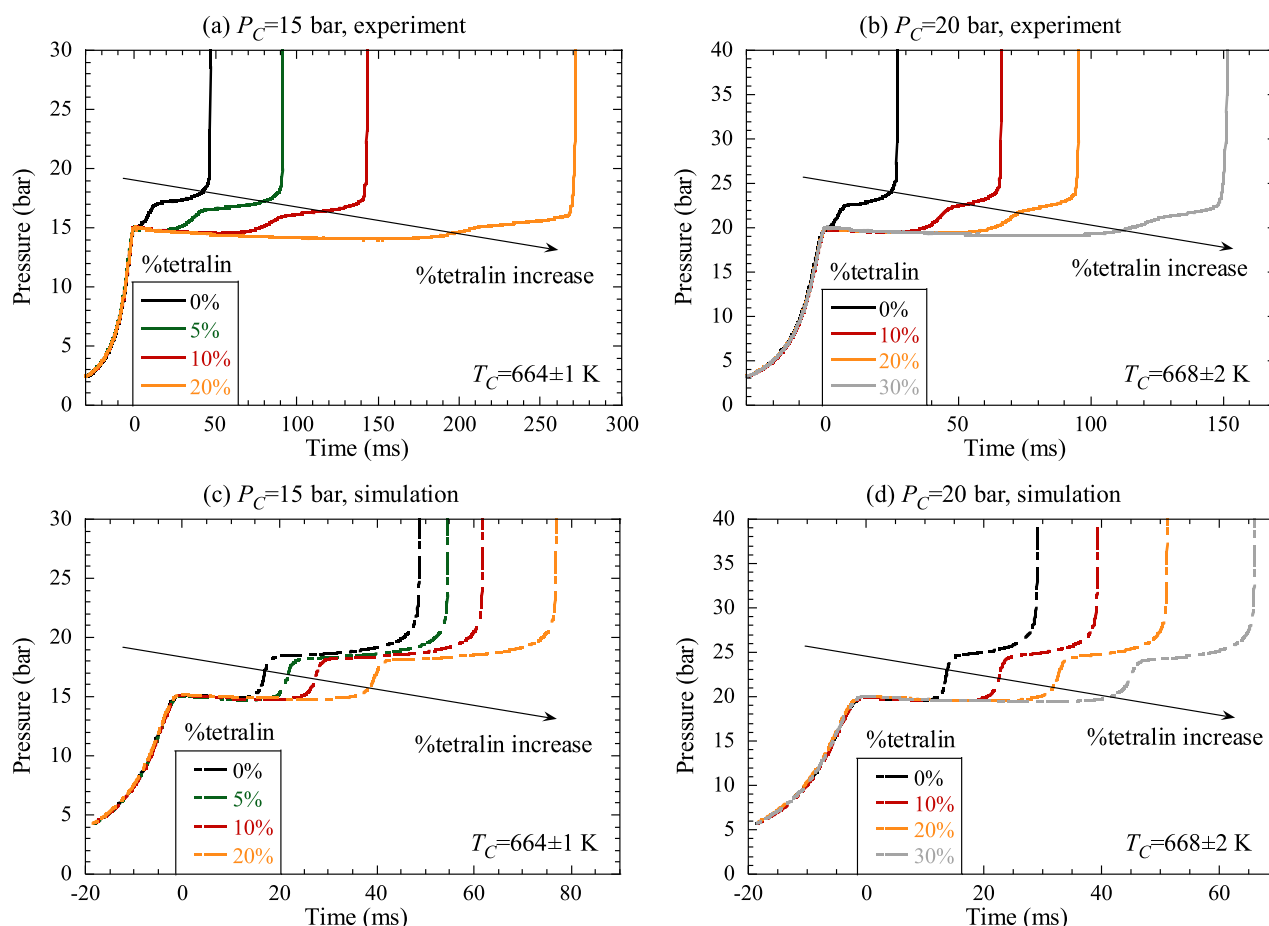


Fig. 5. Comparison of experimental and simulated pressure traces for stoichiometric *iso*-cetane/tetralin blends at varying tetralin values at (a, c) $T_C = 664 \pm 1$ K and $P_C = 15$ bar and (b, d) $T_C = 668 \pm 2$ K and $P_C = 20$ bar.

traces. As clearly seen from Fig. 4, the current model does not capture the effect of increasing tetralin on autoignition data, especially for total and first-stage ignition delay times at temperatures lower than 800 K, with the simulated total ignition delay times being up to a factor of 3 shorter than the experimental data, and the discrepancies between simulated and experimental first-stage ignition results can even be up to a factor of 10. The reactivity over-prediction in total ignition increases with increasing tetralin, which could also be one possible reason causing the over-prediction of the CRC diesel surrogates in the LTR when using the LLNL diesel surrogate model [5], as demonstrated in [4]. Besides, the simulated total ignition delay times of the binary blends are seen to overlap in the ITR, which is the same trend observed in the experimental results. However, the current model predicts longer total ignition delay times than the experimental data in the ITR, which is the similar trend shown in [4] when comparing the experimental and simulated total ignition delay times of the CRC diesel surrogates. As discussed in [4], this total ignition delay time over-prediction in the ITR may be due to the C_0 – C_4 base mechanism from AramcoMech 2.0 [31] that needs further refinements. It is further noted that the current model fails to predict the noticeable increase of total ignition delay times by adding small amount of tetralin in the blend observed in experiments, and the simulated total ignition delay times of neat *iso*-cetane overlap with those of the *iso*-cetane/tetralin blends in the ITR, with the discrepancies between the experimental and simulated results of neat *iso*-cetane being up to a factor of 4. This in turn suggests that the current binary blend model predicts a wider temperature window of the

NTC regime for *iso*-cetane, especially at the higher temperature end when entering the ITR.

Figure 5 plots and compares the experimental and simulated pressure traces of stoichiometric binary blends with varying tetralin values at conditions of $T_C = 664 \pm 1$ K/ $P_C = 15$ bar and $T_C = 668 \pm 2$ K/ $P_C = 20$ bar, which exhibit two-stage ignition characteristics. It is seen from the experimental pressure traces of Figs. 5(a) and 5(b) that the total ignition delay time increases with increasing tetralin (also shown in Fig. 4), which is mostly due to the increase in the first-stage ignition delay time. In addition, the pressure rise associated with the first-stage ignition decreases as tetralin increases, and hence the transition to the second-stage ignition becomes less prominent. Although similar trends can also be observed in Figs. 5(c) and 5(d) for the pressure traces from RCM simulations under the same test conditions, the simulated first-stage pressure rises are higher compared to the experimental results, which could be partially because the multi-dimensional effects of RCM experiments occurring after the first-stage ignition event are not fully captured in the zero-dimensional simulation.

In order to better understand the sensitivity of the total ignition delay time variation with tetralin in the blend, Figs. 6(a) and 6(b) compare the experimental and simulated results of stoichiometric *iso*-cetane/tetralin blends at $P_C = 15$ and 20 bar, as well as at three different compressed temperatures, representing LTR, NTC, and ITR. For the compressed temperatures representing LTR, $T_C = 664 \pm 1$ K in Fig. 6(a) and $T_C = 668 \pm 2$ K in Fig. 6(b), the experimental total ignition delay time increases almost linearly with increasing tetralin in the blend for both compressed pres-

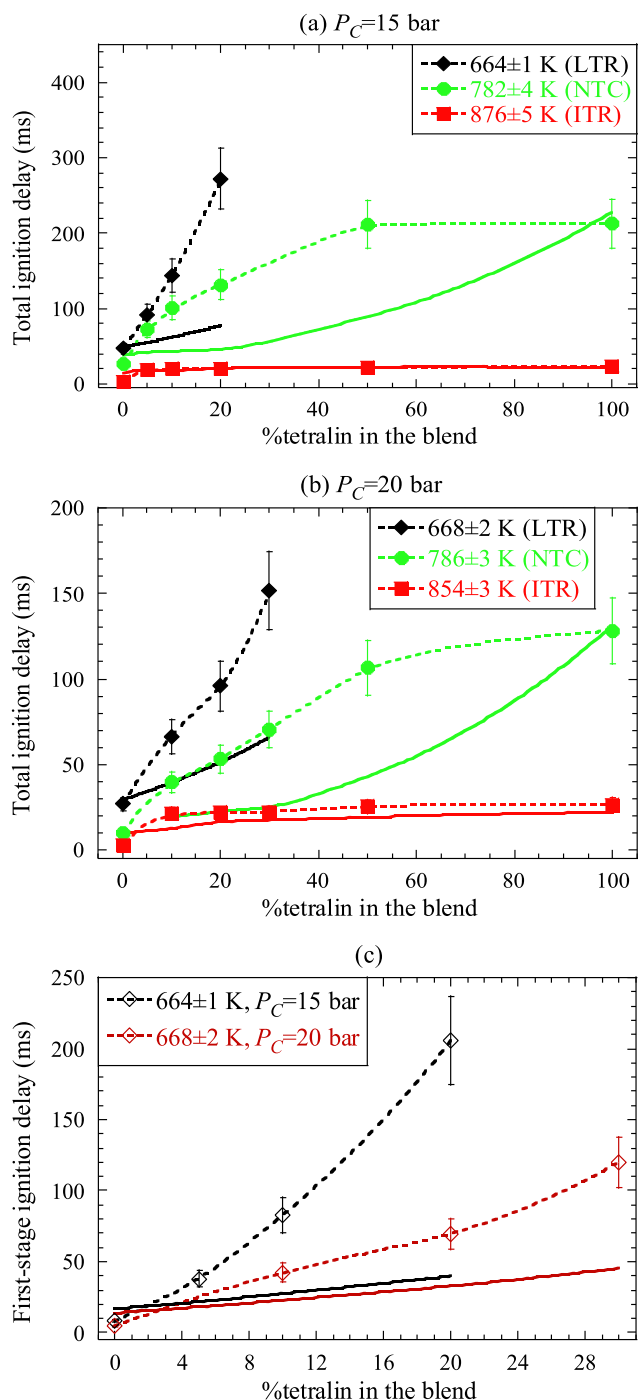


Fig. 6. The effect of tetralin in the stoichiometric *iso*-cetane/tetralin blend on total ignition delay time at (a) $P_C = 15$ bar and (b) $P_C = 20$ bar for three compressed temperatures representing LTR, NTC, and ITR. (c) The effect of tetralin on first-stage ignition delay time at the same LTR conditions in (a) and (b). Symbols with dashed lines: experimental results. Solid lines: simulated results using the current binary blend model.

tures. Specifically, the total ignition delay time increases by a factor of 3 when the tetralin in the blend increases from 0% to 10%, and further increase of tetralin from 10% to 20% also doubles the total ignition delay time at $P_C = 15$ bar. The increase of total ignition delay time by increasing tetralin at $P_C = 20$ bar is less steep compared to that at $P_C = 15$ bar, with the total ignition delay time being doubled as tetralin increases from 0% to 10% and also from 10% to 30%.

Table 2

DCN values of the pure components used in this work and literature binary blend studies [11,34,35]. As the DCN of dimethyl ether is unavailable, its cetane number (CN) is listed instead. DCN₁ and DCN₂ refer to the DCN values of blend species 1 and 2, respectively.

Binary Blend (species 1/species 2)	DCN ₁	DCN ₂
toluene/dimethyl ether [34]	6 [36]	N/A (CN=55 [36])
toluene/ <i>n</i> -heptane [35]	6 [36]	54 [36]
α -methyl-naphthalene/ <i>n</i> -dodecane [11]	0 [2]	73 [36]
α -methyl-naphthalene/isocetane [11]	0 [2]	15.1 [2]
tetralin/ <i>iso</i> -cetane (this work)	8.9 [2]	15.1 [2]

At the compressed temperatures representing the NTC regime for $T_C = 782 \pm 4$ K at $P_C = 15$ bar and $T_C = 786 \pm 3$ K at $P_C = 20$ bar, the dependence of experimental total ignition delay times on the increase of tetralin in the binary blend becomes nonlinear. In particular, increasing tetralin from 0% to 10% in the blends still causes a steep increase (about a factor of 4) in total ignition delay time, but the change in total ignition delay time then becomes less significant (about a factor of 2) by increasing tetralin from 10% to 50%, and further increase of tetralin to 100% does not have much influence on the change of total ignition delay times. At the compressed temperatures representing the ITR, namely $T_C = 876 \pm 5$ K at $P_C = 15$ bar and $T_C = 854 \pm 3$ K at $P_C = 20$ bar, the increase of experimental total ignition delay time is obvious by increasing tetralin from 0% to 5% at $P_C = 15$ bar and from 0% to 10% at $P_C = 20$ bar, but the total ignition delay times remain almost unchanged as tetralin is further increased.

The characteristic response of the current tetralin/*iso*-cetane blends is distinct when comparing with those of toluene/dimethyl ether [34], toluene/*n*-heptane [35], and α -methyl-naphthalene/*n*-dodecane [11]. As discussed in [11,34,35], when two fuel components with distinct reactivities are blended, small addition of the less reactive fuel to the more reactive fuel of abundant presence may have weak or insignificant effect on the overall reactivity of the blend. However, for the tetralin/*iso*-cetane blends show in Fig. 4(a) and 4(b), adding small amount of tetralin (less reactive) to *iso*-cetane (more reactive) can have strong effect in reducing the reactivity of the binary blend. This difference in blending behaviors of the current tetralin/*iso*-cetane mixtures and the literature binary blends of [11,34,35] could be due to the reactivity differences between the fuel components studied. To elaborate, the derived cetane number (DCN) values of the neat fuel components investigated in binary blend studies are taken from [2,36] and compared in Table 2. It can be seen that the difference in ignition propensities of tetralin and *iso*-cetane is much smaller than those of the other three binary blend pairs mentioned above, which could be the possible reason for the distinct blending behavior of tetralin/*iso*-cetane blends. In addition, it is of interest to note from the experimental results of α -methyl-naphthalene/*iso*-cetane blends reported in [11] that adding 25% α -methyl-naphthalene (by mole) to the binary blend has strong effect on the increase of total ignition delay times, but the effect of further increasing α -methyl-naphthalene is relatively less pronounced. As shown in Table 2, the difference in DCN values of α -methyl-naphthalene and *iso*-cetane is within the similar order of magnitude as the DCN difference between tetralin and *iso*-cetane (15.1 vs. 6.2, respectively), thereby both binary blends exhibiting similar blending behavior.

As demonstrated earlier, the current model fails to capture the effect of increasing tetralin on total ignition delay time, especially at temperatures lower than 800 K, which can also be clearly observed from the simulated results shown in Figs. 6(a) and 6(b). In the LTR, while the simulated results capture the linearly increasing trend of the experimental results when adding tetralin in the blends, the under-prediction of total ignition delay times

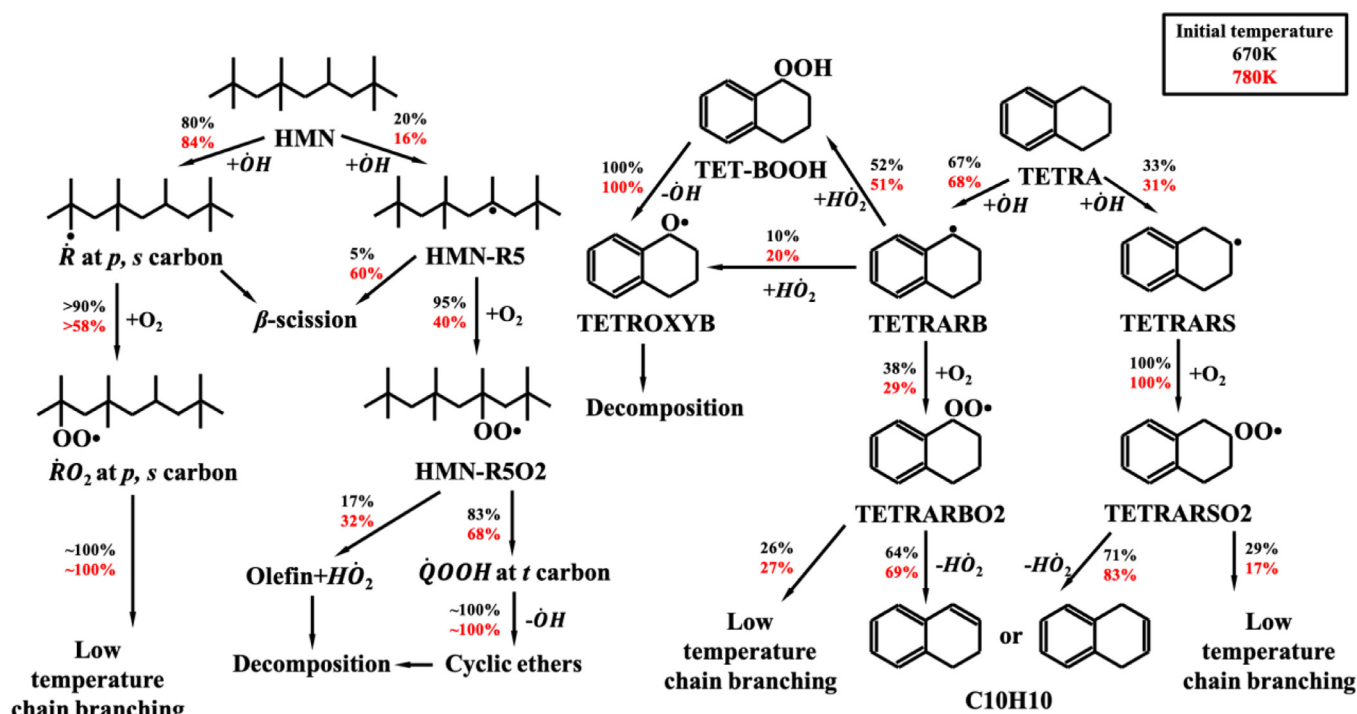


Fig. 7. Reaction pathway analysis showing the important fuel consumption pathways of HMN and TETRA in the stoichiometric 20/80 TETRA/HMN blend at the time of 10% consumption of HMN. CONV simulations are conducted at initial pressure of 15 bar and initial temperatures of 670 K and 780 K.

can be up to a factor of 3. Comparison of experimental and simulated first-stage ignition delay times at temperatures representing LTR at $P_C = 15$ and 20 bar has also been shown in Fig. 6(c), and this comparison illustrates the same trend as that of total ignition delay times. Specifically, both experimental and simulated results increase linearly with increasing %tetralin, but the current model over-predicts the reactivities of binary blends at both pressures, with discrepancies being up to a factor of 5 at $P_C = 15$ bar and up to a factor of 2.5 at $P_C = 20$ bar. This comparison further indicates that the discrepancies in first-stage ignition predictions can be one possible reason causing the poorer agreements of total ignition delay times in the LTR. Another probable reason for the observed discrepancy is the difference in heat release rate from the first-stage ignition between experiment and simulation. As seen in Fig. 5 for the pressure trace comparison, the current model predicts higher level of heat release rate as indicated by the higher pressure rise from the first-stage ignition, which can lead to shorter simulated total ignition delay times with %tetralin increasing from 5% to 20% at $P_C = 15$ bar and from 10% to 30% at $P_C = 20$ bar.

In the NTC regime, the current model cannot even capture the trend of experimental data with the increase of %tetralin, which is also shown in Figs. 4(a) and 4(b). In particular, the experimental results show that the effect of adding %tetralin on total ignition delay time becomes gradually insignificant as %tetralin increases, while the simulated results display a completely opposite trend, with the simulated total ignition delay times increase continuously with the increase of %tetralin. In addition, the noticeable increase of experimental total ignition delay times by adding small amount of tetralin especially in the NTC regime and ITR is also not well-predicted by the current model. It has been shown in Fig. 2 that the simulated total ignition delay times of neat *iso*-cetane agree reasonably well with the experimental results in the LTR and NTC regime, and the reactivity of tetralin has also been reasonably predicted by the current model. Since both Figs. 4 and 6 demonstrate that the current model does not capture the increase trend of total ignition delay times by adding %tetralin, it is of impor-

tance to examine whether the inclusion of the chemical kinetic interactions between two fuel components can improve the model performance. Thus, chemical kinetic analyses are conducted and discussed in next section to identify the key reactions governing the autoignition chemistry of binary blends based on the current model, and the effect of cross-reactions on autoignition prediction of *iso*-cetane/tetralin blends will be delineated in Section 4.4.

4.3. Chemical kinetic analysis results

Reaction pathway analysis has been conducted to help identify the important reaction process and understand the chemistry responsible for blending behavior, and the results are shown in Fig. 7. In order to be consistent with species names in the current binary blend model and with the study of [4], HMN and TETRA are used to denote *iso*-cetane and tetralin, respectively, starting from this section. Moreover, the present reaction pathway analysis has been performed using CONV simulations at the time of 10% HMN fuel consumption (the associated temperature rise is ~9 K above the initial temperature) for a stoichiometric 20/80 TETRA/HMN (20% tetralin and 80% *iso*-cetane by mole) binary blend at initial pressure of 15 bar and initial temperatures of 670 K and 780 K, which are representative temperatures in the LTR and NTC regime, respectively.

It is seen from Fig. 7 that both HMN and TETRA are predominantly consumed by H-abstraction of OH as the first step of reactions. For HMN, eight types of radicals (HMN-R1 to HMN-R8) can be produced from H-abstraction of OH, with H radical abstracted from primary (*p*), secondary (*s*), and tertiary (*t*) carbons. A sketch showing site numbers of HMN is displayed in Fig. S1 in Supplementary Material. These radicals then participate in the O_2 addition process and form peroxy adducts, with higher percentages at the lower temperature of 670 K, while the rest enter the β -scission pathways and the subsequent isomerization to other HMN radicals (not shown in Fig. 7). All RO_2 radicals from O_2 addition at *p* and *s* carbons then initiate the low temperature chain branching re-

actions. Most RO_2 radicals from O_2 addition at t carbons (HMN-R5O2) isomerize to QOOH radicals which then form cyclic ethers and OH radicals, and the rest RO_2 radicals participate in the concerted elimination pathways producing olefins and hydroperoxyl radicals (HO_2).

For TETRA, two types of tetralyl radicals, TETRARS and TETRARB, are formed predominantly from H-abstraction reactions by OH radicals. For both temperatures, all TETRARS and less than 40% of TETRARB are consumed by reactions with O_2 forming tetralyl peroxy radicals (TETRARSO2 and TETRARB02), and the majority of tetralyl peroxy radicals undergo concerted elimination reactions to form dialin ($\text{C}_{10}\text{H}_{10}$) and HO_2 radicals. While not shown in Fig. 7, most $\text{C}_{10}\text{H}_{10}$ at both temperatures participate in the H-abstraction reactions by OH and form C_{10}H_9 radicals that further decompose to smaller radicals. The rest majority of TETRARB radicals become TETROXYB via reactions with HO_2 radicals, and TETROXYB radicals further enter the decomposition pathways.

In a study on dimethyl ether/propane blended oxidation, Dames et al. [37] showed that the important interactions of a binary fuel blend with high- and low-reactivity are through OH and HO_2 radicals instead of cross-reactions between two fuels and their fragments. In addition, since nearly all HMN and TETRA are consumed by H-abstraction of OH as the first step of reactions in the test conditions of Fig. 7 and since these reactions are dominant in controlling the reactivity as seen from the sensitivity analysis results of the CRC surrogates [4] and of the current binary blends (shown later), a rate of destruction (ROD) analysis of OH radicals is conducted next for stoichiometric *iso*-cetane/tetralin blends at varying blending ratios. Similar to the above reaction pathway analysis, the ROD analysis is carried out via CONV simulations at the time of 10% HMN consumption with initial pressure of 15 bar and initial temperatures of 670 K and 780 K, respectively representing the LTR and NTC regime. Here, the OH consumption ratio is defined by:

$$\text{OH consumption ratio} = \frac{\sum \text{ROD for H-abstraction of TETRA by OH}}{\sum \text{ROD for H-abstraction of HMN by OH}},$$

which provides information on the consumption of OH radicals by TETRA relative to that by HMN. The OH consumption ratio, as well as the total ignition delay times from RCM experiments and from CONV simulations, as a function of %TETRA in the stoichiometric blend at 670 K and 780 K are plotted in Fig. 8.

It is seen from Figs. 8(a) and 8(b) that while both experiment and simulated total ignition delay times increase in the %TETRA range of 0–20% at temperatures representing LTR and NTC, the blend reactivity is over-predicted and the slope of the experimental data is not captured by CONV simulations (also cf. Fig. 6(a) for not being captured by RCM simulations either). On the other hand, the increasing trends of the simulated OH consumption ratio with increasing %TETRA from 0% to 20% appear to follow the increasing experimental total ignition delay time trends at both temperatures. However, when covering the entire %TETRA range from 0% to 100% at the NTC-relevant temperature (~780 K), Fig. 8(c) shows that the experimental results exhibit a concave increase with increasing %TETRA and then level off beyond 50% blending, but both the simulated ignition delay times and OH consumption ratios display a convex increase, meaning that their increasing trends become steeper with the increase of %TETRA. As a result, it can be concluded that the simulated ignition delay times and OH consumption ratios follow the same increasing trend with increasing %TETRA in the LTR and NTC regimes, which is consistent with the reaction pathway analysis results demonstrated in Fig. 7 that both fuel components in the blend are predominantly consumed by H-abstraction of OH. Since low temperature chemistry of fuel radicals produced from H-abstraction typically governs the overall reactivity, this further indicates that the discrepancies of blending behavior between experimental and simulated ignition delay

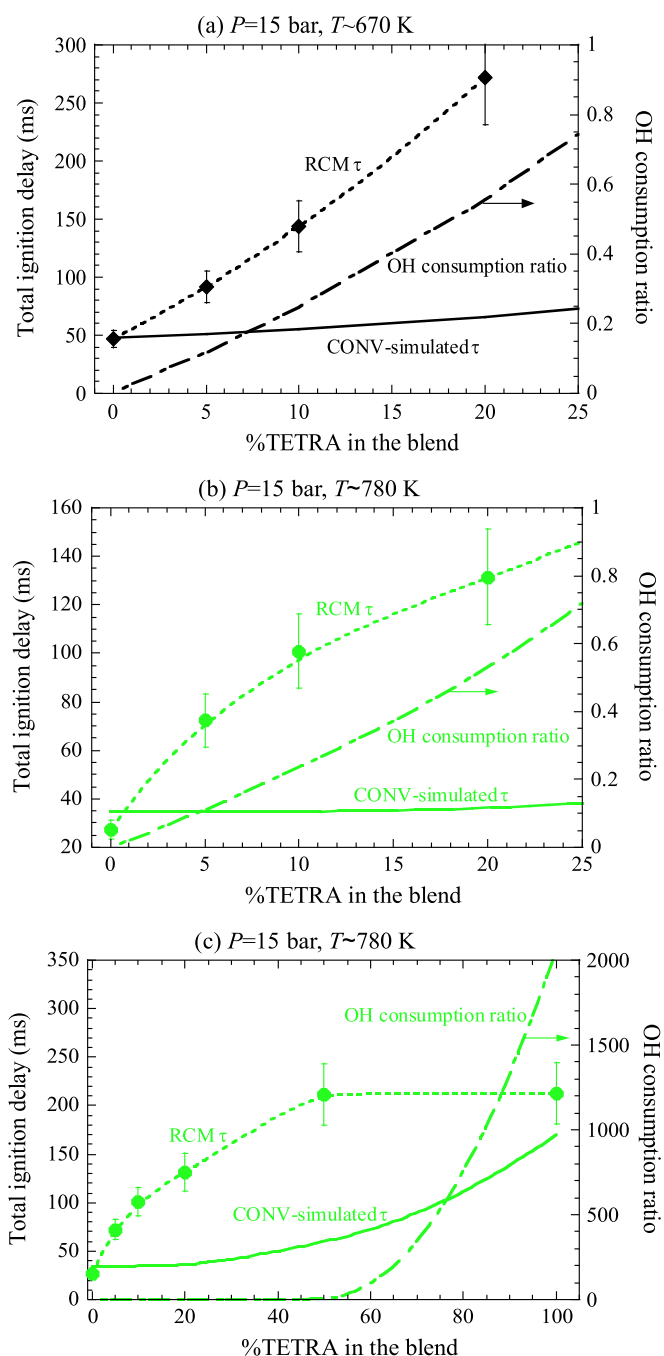


Fig. 8. The effect of %TETRA in stoichiometric HMN/TETRA blends on simulated OH consumption ratio and CONV-simulated total ignition delay time at initial pressure of 15 bar and initial temperatures of 670 K and 780 K. The present RCM total ignition delay times at $P_C=15$ bar and $T_C=664$ K and ~ 782 K are also shown for comparison. Fig. 8(a) shows the results at ~ 670 K, and Figs. 8(b) and 8(c) show the results at ~ 780 K with %TETRA up to 25% and 100%, respectively.

times in the LTR and NTC regime may be related to the inaccurate OH consumption proportions between HMN and TETRA. Therefore, by improving the prediction of OH consumption distributions, the current model may be able to better predict the effects of increasing %TETRA on autoignition at low temperatures and the noticeable increase of experimental total ignition delay times by increasing %TETRA from 0% to 5% at $P_C = 15$ bar and from 0% to 10% at $P_C = 20$ bar, as respectively shown in Figs. 4(a) and 4(b).

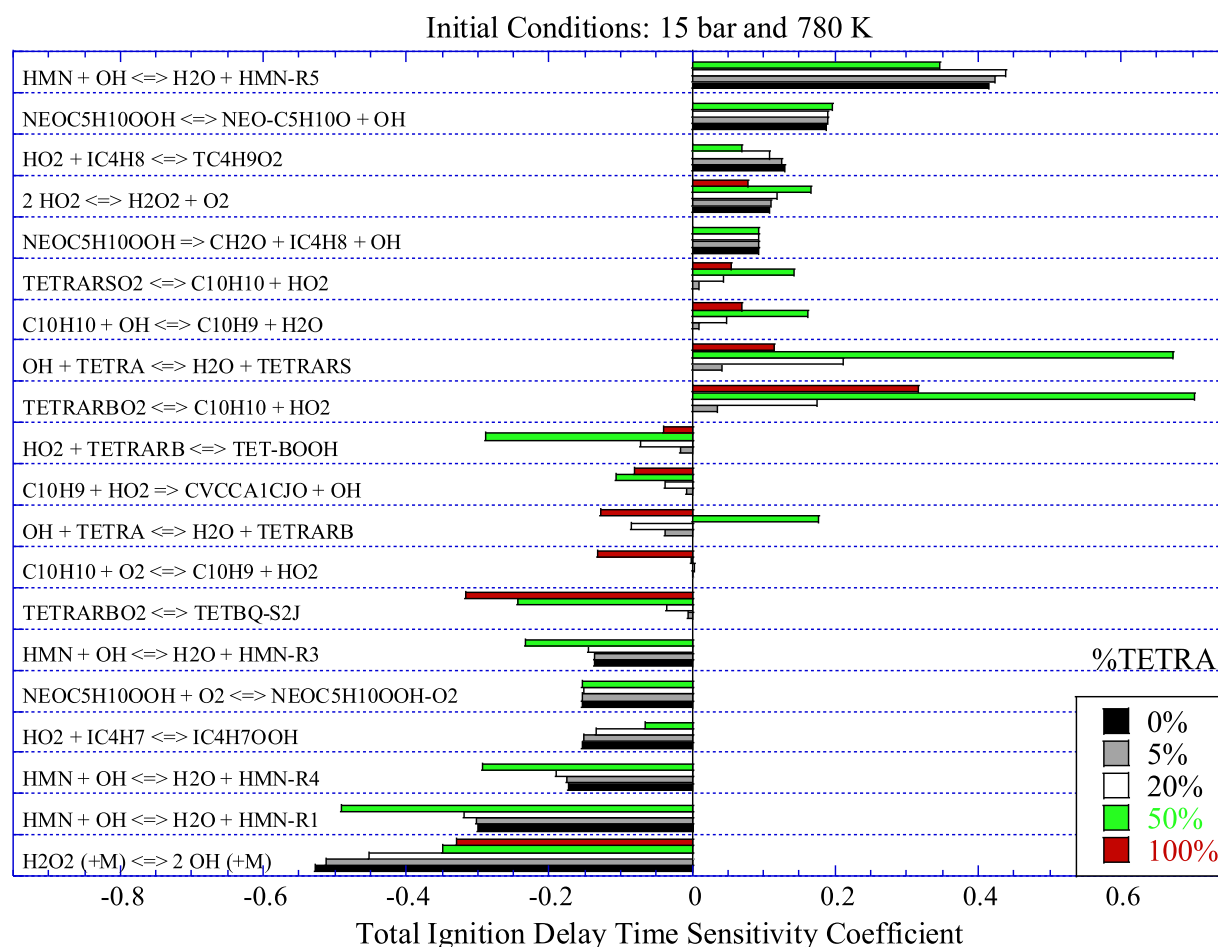


Fig. 9. Total ignition delay time sensitivity coefficients of important promoting/inhibiting reactions at different blending ratios in stoichiometric HMN/TETRA blends with initial pressure of 15 bar and initial temperature of 780 K obtained from CONV simulations.

In order to further understand the discrepancies between experimental and simulated results at low temperatures when increasing %TETRA in the binary blend, brute force sensitivity analyses to identify the controlling chemistry for autoignition have been performed under CONV simulations based on the current model. The total ignition delay time sensitivity coefficient is defined as $S_{rel} = \ln(\tau_+/\tau)/\ln(k_+/k)$, where k and τ are the unperturbed reaction rate coefficient and the corresponding total ignition delay time, respectively, and k_+ is the rate coefficient increased by a factor of two while τ_+ is the resulting total ignition delay time. Therefore, a reaction with a positive (negative) S_{rel} inhibits (promotes) the overall reactivity.

Figure 9 shows the sensitivity coefficient results for stoichiometric binary blends at initial pressure of 15 bar and initial temperature of 780 K (representing the NTC regime) with selected blending ratios to highlight the change of important reactions that affect the predicted reactivity when increasing %TETRA. It can be observed that the H-abstraction reactions of HMN and TETRA by OH are among important reactions controlling the reactivity for the blending ratios investigated, which is consistent with the previous discussion. For HMN, the sensitivity coefficients of those reactions exhibit site specificity. For instance, the H-abstraction reaction for HMN leading to HMN-R5 inhibits the reactivity, which is because QOOH radicals on t carbons cannot further produce ketohydroperoxides (KHPs) through the subsequent propagation reactions. As shown in Fig. 7, QOOH radicals on t carbons produce cyclic ethers instead. Moreover, the H-abstraction reactions of TETRA by OH producing TETRARS and TETRARB are shown in Fig. 9 to inhibit the

reactivity for certain blending ratios. This is because these tetralyl radicals can form tetralyl peroxy radicals by reacting with O₂ and the majority of the formed tetralyl peroxy radicals subsequently participate in the concerted elimination reactions forming dialin (C₁₀H₁₀) and HO₂ that inhibit reactivity, which can also be seen in Fig. 7. Figure S2 in Supplementary Material further shows the total ignition delay time sensitivity coefficient variation of the reaction of OH+TETRA $\rightleftharpoons$ H₂O+TETRARB as a function of %TETRA in the blend at the same initial conditions as in Fig. 9, by reporting the sensitivity coefficients at an increment of 10% TETRA in the blend. It is demonstrated that the variation in sensitivity coefficients change smoothly from negative to positive with the increase of %TETRA, and then become negative again when %TETRA reach 80%. In addition, a reaction pathway analysis of TETRA in the stoichiometric 50/50 TETRA/HMN blend at the time of 10% consumption of HMN is shown in Fig. S3 in Supplementary Material, with the initial conditions of 15 bar and 780 K. As compared to the consumption of TETRA for the stoichiometric 20/80 TETRA/HMN blend at 780 K shown in Fig. 7, the 50/50 TETRA/HMN blend is seen to have a much higher percentage of TETRARB entering the O₂ addition pathway producing TETRARBO₂ (48% vs. 29%), and this increase can in turn lead to more production of C₁₀H₁₀ that inhibits the reactivity. This is also consistent with the positive sensitivity coefficient of OH+TETRA $\rightleftharpoons$ H₂O+TETRARB at 50%TETRA as observed in Fig. 9 and Fig. S2.

As TETRA is added into the blend from 0% to 5%, the top-ranked promoting and inhibiting reactions remain unchanged by comparing to the case with neat HMN (0%TETRA). By further

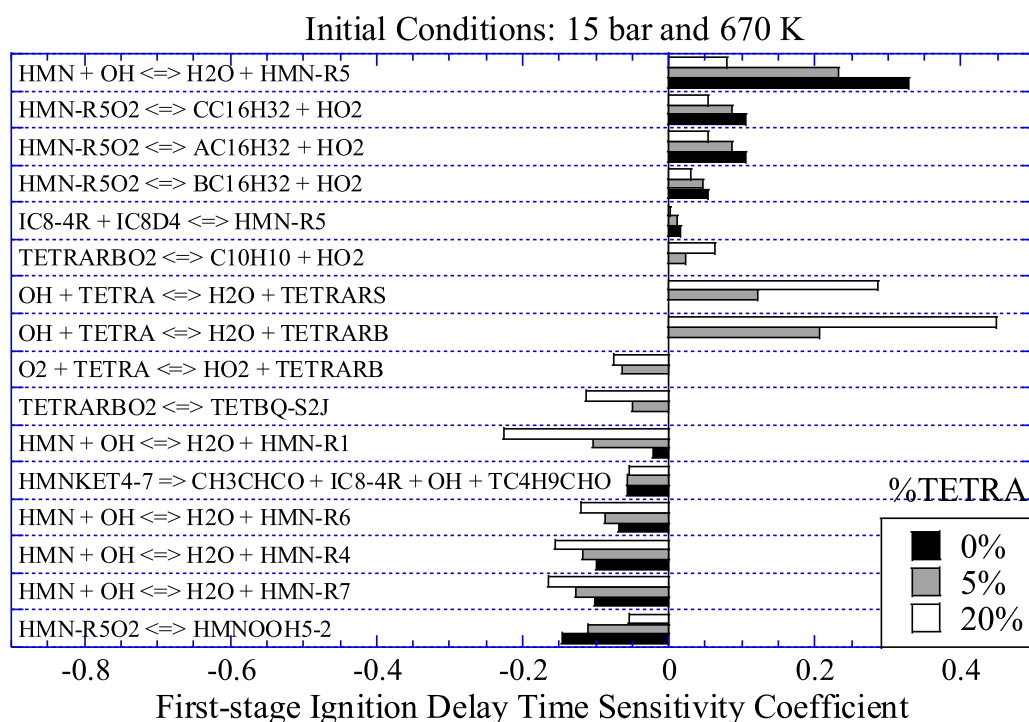


Fig. 10. First-stage ignition delay time sensitivity coefficients of important promoting/inhibiting reactions at different blending ratios in stoichiometric HMN/TETRA blends with initial pressure of 15 bar and initial temperature of 670 K obtained from CONV simulations.

increasing%TETRA in the blend, an increasing dominance of the TETRA-related reactions can be seen in Fig. 9. It is also noted from Fig. 9 that the TETRA-related reactions are not among the top-ranked reactions controlling the reactivity at lower%TETRA but their dominance increases substantially with further increasing%TETRA that may lead to the convex increase of simulated total ignition delay times shown in Fig. 6, which is of the opposite trend comparing to the experimental total ignition delay times. In addition, as%TETRA increases from 0% to 5%, most of the top-ranked dominant reactions are related to HMN. This is also in contradict with the experimental results that adding small amount of TETRA to the blend has strong effect on total ignition delay time as mentioned previously. Therefore, the relative dominance between the reactions of TETRA and HMN may need refinement to have the TETRA-related reactions to be more dominant at lower%TETRA and less dominant at higher%TETRA, in order to better match the experimental trend.

Using the same sensitivity coefficient definition, sensitivity analyses for first-stage ignition delay times are conducted as well. It is seen from Figs. 4 and 5 that the increase in total ignition delay time with increasing%TETRA is mainly due to the increase of first-stage ignition delay time in both experiment and simulation. Besides, Fig. 6(c) suggests that the discrepancies in first-stage ignition delay time predictions can be one possible reason causing the poor agreements of total ignition delay times in the LTR. To identify the dominant reactions controlling the reactivity of first-stage ignition in the LTR, Fig. 10 shows the sensitivity analysis results for selected blending ratios in stoichiometric HMN/TETRA blends at initial pressure of 15 bar and initial temperatures of 670 K. For neat HMN, it can be observed that the dominant reactions promoting the reactivity include the isomerization reaction of HMN-R5O2 (RO₂ radical) forming HMNOOH5-2 (QOOH radical) and H-abstraction of HMN by OH at *p* and *s* carbons producing HMN-R4, -R6, and -R7 radicals (site numbers can be seen in Fig. S1 in Supplementary Material). Moreover, H-abstraction of HMN by OH at *t* carbon producing HMN-R5 is still the top re-

action inhibiting the reactivity, and the concerted elimination reactions of HMN-R5O2 are the other dominant reactions inhibiting the reactivity. Those HMN-related reactions remain important when adding%TETRA to the blend. Among the TETRA-related reactions, the isomerization reaction from TETRARBO2 to TETBQ-S2J (QOOH radical) promotes reactivity, while H-abstraction of TETRA by OH and concerted elimination of TETRARBO2 inhibit reactivity. In addition, these TETRA-related reactions gradually become top-dominant reactions as%TETRA increases. In order to address the issues of the current model prediction in the LTR, various TETRA-related reactions can be revisited in concert to achieve better agreements with the experimental results at different blending ratios, and the dominant HMN-related reactions can also be refined for better agreements with the neat HMN data. Furthermore, Fig. S4 in Supplementary Material shows the sensitivity coefficient results of neat HMN at initial pressure of 15 bar and initial temperature of 670 K (representing LTR). According to the current model predictions of neat HMN in Fig. 2, in order to improve the prediction of first-stage ignition delay times in the LTR without affecting the performance of total ignition delay time predictivity, one may focus more on refining the rate coefficients of the reactions that are more dominant in first-stage ignition but not in total ignition, such as the decomposition reactions of KHP and the isomerization and concerted elimination reactions of HMN-R5O2. It is also seen from Fig. S4 that the H-abstraction reactions of HMN by OH radicals are dominant in controlling both first-stage and total ignition delay times, and this further indicates the importance of accurate prediction of OH consumption by neat fuels.

4.4. Effect of cross-reactions on autoignition prediction

As the current binary blend model does not include any cross-reactions among HMN, TETRA, and their fragments, and as the simulated ignition delay times are unable to fully capture the experimental trends of HMN/TETRA blends especially at temperatures below 800 K, whether including those cross-reactions can help im-

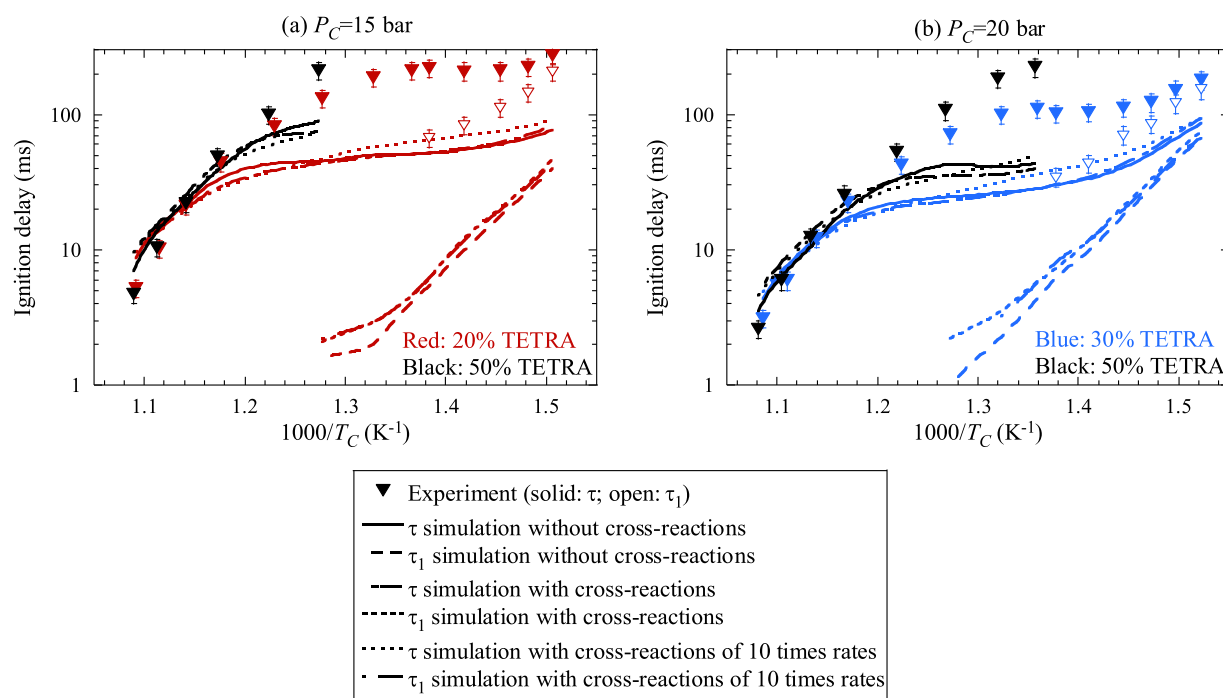


Fig. 11. Comparison of simulated total (τ) and first-stage (τ_1) ignition delay times using the current model with and without cross-reactions for stoichiometric HMN/TETRA blends with selected blending ratios at (a) $P_C = 15$ bar and (b) $P_C = 20$ bar. Experimental data are also included as a reference.

prove the prediction of the blending effects is explored in the following. Here, the cross-reactions added to the current model are among HMN, TETRA, their radicals from H-abstraction, and RO_2 radicals, with their rate parameters taken from the analogous reactions in Mehl et al. [38]. The details of those cross-reactions and their corresponding sources of rate parameters can be found in Supplementary Material.

RCM simulations have been conducted and compared with and without the above-mentioned cross-reactions in order to illustrate the effect of those cross-reactions on simulated ignition delay times. The comparisons are shown in Fig. 11 for stoichiometric 20/80 and 50/50 TETRA/HMN blends at $P_C = 15$ bar, as well as stoichiometric 30/70 and 50/50 TETRA/HMN blends at $P_C = 20$ bar, along with experimental ignition delay times as a reference. It can be observed that by including cross-reactions, the simulated total ignition delay times of the 20/80 and 30/70 TETRA/HMN blends increase slightly (less than 9%) compared to the simulated results without cross-reactions at temperatures lower than 735 K, and decrease slightly (less than 15%) in the temperature range of 735–885 K. Moreover, as no literature study has been conducted on the rate parameters of those cross-reactions of the current interest to the best of our knowledge, their estimated rates need further scrutiny. Therefore, in order to demonstrate the sensitivity of the simulation results to the rate parameter uncertainties in these cross-reactions, all rates of the included cross-reactions are further increased by a factor of 10, and such simulated results have also been shown in Fig. 11. By further increase the rates of cross-reactions by a factor of 10, the increase of the simulated total ignition delay times is less than 30% at temperatures below 800 K for the 20/80 and 30/70 TETRA/HMN blends. For the simulated first-stage and total ignition delay time results at other temperatures and test conditions, the effects of including cross-reactions with $10 \times$ rates seem to be similar to those with cross-reactions of their original rates. Namely, the simulated first-stage ignition delay times for the 20/80 and 30/70 TETRA/HMN blends are observed to have average of 15% increase by including cross-reactions

of either rates. In addition, for the 50/50 TETRA/HMN blends at both $P_C = 15$ and 20 bar, the simulated total ignition delay times with cross-reactions of either rates are observed to have slight decrease in the temperature range of 770–830 K and slight increase at temperatures higher than 830 K, with most changes being less than 20%. As discussed earlier, the simulated total and first-stage ignition delay times at temperatures below 800 K can be respectively up to a factor of 3 and 10 shorter than the experimental data. Hence, it can be concluded that adding those cross-reactions to the current model only yields marginal improvement in the prediction, and they play minimal roles in accounting for the observed discrepancies between experimental and simulated results in the LTR and NTC regime. This is consistent with the statements from Dames et al. [37] that it is not necessary to consider cross reactions of fuel fragments during combustion of high-/low-reactivity fuel blends, and that simple addition and de-duplication of separately validated models is sufficient to produce a combined model.

In view of the above analyses in both Sections 4.3 and 4.4, it is expected that the main chemical interactions between *iso*-cetane and tetralin are through small radical species like OH and HO_2 , instead of their cross-reactions. In particular, accurate prediction of OH consumption distribution between the two fuel components may hold the key to major improvement of the current model. The newly-acquired RCM results herein represent targets for developing/refining the next generation of chemical kinetic models that will require more accurate rate constants and thermodynamic properties. Thus, a comprehensive review of the base chemical kinetic models of *iso*-cetane and tetralin is critically needed in order to help explain the underlying chemistry behind the observed blending behaviors. It has to be pointed out that the importance of thermodynamic properties cannot be over-emphasized. Again, the comparison of experimental and simulated first-stage ignition delay times indicates that the larger discrepancies in first-stage ignition delay times typically result in the larger discrepancies in total ignition delay times. Those discrepancies can be decreased by first investigating the thermochemistry of *iso*-cetane at low-to-

intermediate temperatures, and subsequently further rate parameter refinements for the reactions controlling the reactivity can be carried out more effectively. As shown in Mehl et al. [39], the revision of thermodynamic properties of several radicals including iso-octyl isomers and neo-pentyl is an important improvement to their gasoline surrogate model. In addition, Bugler et al. [40] demonstrated the significant change of reactivity at temperatures lower than 1000 K by updating the thermochemistry of several classes of C_5 species that are important in low-temperature oxidation, and hence concluded that revisions to both the thermochemistry and the kinetics are required in order to replicate experiments well. Recognizing that model development is an evolutionary process, once more accurate thermodynamic properties and rate constants become available, their incorporation into the sub-models shall make the surrogate models more predictive.

5. Conclusions

Autoignition of *iso*-cetane, tetralin, and their binary blends has been studied in a rapid compression machine at low-to-intermediate temperatures with varying compressed pressures and blending ratios. The current RCM ignition delay times of neat components are seen to have reasonable agreement with the literature RCM data. In addition, the ignition delay time data of two pure fuels and their binary blends have been compared with the simulated results using a chemical kinetic model assembled from the LLNL diesel surrogate model with some updates. It is found that the simulated results have good agreement with the experimental data of tetralin. For *iso*-cetane, although the current model has reasonable predictions on ignition delay times at lower temperatures, discrepancies between experiments and simulations become larger as temperature increases. In addition, the current model fails to fully capture the obvious increase of total ignition delay times by adding small amount of tetralin in the blend, especially in the NTC regime and ITR, and it cannot predict the increase of total ignition delay time by further increasing tetralin in the LTR and NTC regime, which are the major issues of the current model.

Chemical kinetic studies of the binary blends, including the reaction pathway, OH consumption ratio, and brute force sensitivity analyses, have been conducted at varying pressures, temperatures, and blending ratios to understand the discrepancies between the experimental and simulated results. It is demonstrated that the discrepancies may be related to the inaccurate prediction of OH consumption distribution between the two parent fuels. Sensitivity analysis results also help identify the important reactions governing autoignition at different blending ratios and provide guidance to the potential refinements on the current model to improve its prediction. Furthermore, the effect of cross-reactions between the two fuels and their fragments have also been investigated by comparing simulated results with and without including those cross-reactions. In order to elucidate the influence of rate parameter uncertainties on the simulated results, the rates of all the cross-reactions considered are further increased by a factor of 10. It can be concluded that including those cross-reactions only has slight influence on the simulated results, and the major improvement of the current model still relies on accurate prediction of OH consumption distribution between *iso*-cetane and tetralin. Through the present investigation, the refinements and development of comprehensive diesel surrogate models can be benefited by understanding the autoignition behavior of *iso*-cetane/tetralin blends at low-to-intermediate temperatures.

Supplementary material

Supplementary Material consists of (1) Figures S1–S4, (2) a summary of the current ignition delay time data, (3) the de-

tailed chemical kinetic model for *iso*-cetane/tetralin blends, (4) cross-reactions between fuels and their fragments considered in Section 4.4, and (5) RCM experimental pressure traces and the deduced time-volume histories for RCM simulations, which are available from the website of the Combustion Diagnostics Laboratory at UCONN (<https://combdialab.engr.uconn.edu/database/rcm-database>).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

Acknowledgment

The work at UCONN was supported by LLNL via Standard Research Subcontract No. B630575. The work at LLNL was performed under the auspices of the U.S. Department of Energy (DOE) by Lawrence Livermore National Laboratory under Contract DE-AC52-07NA27344 and was conducted as part of the Co-Optimization of Fuels & Engines (Co-Optima) project sponsored by the DOE Office of Energy Efficiency and Renewable Energy (EERE), Bioenergy Technologies and Vehicle Technologies Offices (program managers: Gurpreet Singh and Kevin Stork).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2021.02.018.

References

- [1] M.P.B. Musculus, P.C. Miles, L.M. Pickett, Conceptual models for partially pre-mixed low-temperature diesel combustion, *Prog. Energy Combust. Sci.* 39 (2013) 246–283.
- [2] C.J. Mueller, W.J. Cannella, J.T. Bays, T.J. Bruno, K. DeFabio, H.D. Dettman, R.M. Gieleciak, M.L. Huber, C.-B. Kweon, S.S. McConnell, W.J. Pitz, M.A. Ratcliff, Diesel surrogate fuels for engine testing and chemical-kinetic modeling: compositions and properties, *Energy Fuels* 30 (2016) 1445–1461.
- [3] C.J. Mueller, W.J. Cannella, T.J. Bruno, B. Bunting, H.D. Dettman, J.A. Franz, M.L. Huber, M. Natarajan, W.J. Pitz, M.A. Ratcliff, K. Wright, Methodology for formulating diesel surrogate fuels with accurate compositional, ignition-quality, and volatility characteristics, *Energy Fuels* 26 (2012) 3284–3303.
- [4] M. Wang, G. Kukkadapu, K. Zhang, S.W. Wagnon, M. Mehl, W.J. Pitz, C.K. Westbrook, C.J. Sung, Autoignition of CRC diesel surrogates at low temperature combustion conditions: rapid compression machine experiments and modeling, *Combust. Flame* 219 (2020) 178–197.
- [5] G. Kukkadapu, R. Whitesides, M. Wang, S.W. Wagnon, K. Zhang, M. Mehl, C.K. Westbrook, R. McCormick, C.J. Sung, W.J. Pitz, Experimental and surrogate modeling study of diesel fuel, in preparation; see preliminary version in, 11th Joint Meeting of the U.S. Sections of the Combustion Institute, Paper No. 1B14, Pasadena, CA, 2019.
- [6] M.A. Oehlschlaeger, J. Steinberg, C.K. Westbrook, W.J. Pitz, The autoignition of *iso*-cetane at high to moderate temperatures and elevated pressures: shock tube experiments and kinetic modeling, *Combust. Flame* 156 (2009) 2165–2172.
- [7] P. Dagaut, K. Hadj-Ali, Chemical kinetic study of the oxidation of *iso*-cetane in a jet stirred reactor: experimental and modeling, *Energy Fuels* 23 (2009) 2389–2395.
- [8] B. Li, H. Zhang, F.N. Egolfopoulos, Laminar flame propagation of atmospheric *iso*-cetane/air and decalin/air mixtures, *Combust. Flame* 161 (2014) 154–161.
- [9] S.H. Won, S. Dooley, P.S. Veloo, H. Wang, M.A. Oehlschlaeger, F.L. Dryer, Y. Ju, The combustion properties of 2,6,10-trimethyl dodecane and a chemical functional group analysis, *Combust. Flame* 161 (2014) 826–834.
- [10] S.H. Won, F.M. Haas, A. Tekawade, G. Kosiba, M.A. Oehlschlaeger, S. Dooley, F.L. Dryer, Combustion characteristics of C4 *iso*-alkane oligomers: experimental characterization of *iso*-dodecane as a jet fuel surrogate component, *Combust. Flame* 165 (2016) 137–143.
- [11] G. Kukkadapu, C.J. Sung, Autoignition study of binary blends of *n*-dodecane/1-methylnaphthalene and *iso*-cetane/1-methylnaphthalene, *Combust. Flame* 189 (2018) 367–377.

- [12] L. Yu, Y. Qiu, Y. Mao, S. Wang, C. Ruan, W. Tao, Y. Qian, X. Lu, A study on the low-to-intermediate temperature ignition delays of long chain branched paraffin: Iso-cetane, *Proc. Combust. Inst.* 37 (2019) 631–638.
- [13] Y. Yang, A.L. Boehman, Oxidation chemistry of cyclic hydrocarbons in a motored engine: methylcyclopentane, tetralin, and decalin, *Combust. Flame* 157 (2010) 495–505.
- [14] H. Wang, W.J. Gerken, W. Wang, M.A. Oehlschlaeger, Experimental study of the high-temperature autoignition of tetralin, *Energy Fuels* 27 (2013) 5483–5487.
- [15] P. Dagaut, A. Ristori, A. Frassoldati, T. Faravelli, G. Dayma, E. Ranzi, Experimental study of tetralin oxidation and kinetic modeling of its pyrolysis and oxidation, *Energy Fuels* 27 (2013) 1576–1585.
- [16] G. Kukkadapu, B.W. Weber, C.J. Sung, Autoignition study of tetralin in a rapid compression machine at elevated pressures and low-to-intermediate temperatures, *Fuel* 159 (2015) 436–445.
- [17] M. Raza, Y. Qian, S. Wang, Y. Mao, J. Zhu, X. Lu, The experimental study of autoignition of tetralin at intermediate-to-high temperatures, *Fuel* 266 (2020) 117081.
- [18] R.D. Wilk, *Preignition Oxidation Characteristics of Hydrocarbon Fuels*, Drexel University, Philadelphia, PA, 1986 <https://www.osti.gov/biblio/7148186>.
- [19] G. Mittal, C.J. Sung, A rapid compression machine for chemical kinetic studies at elevated pressure and temperatures, *Combust. Sci. Technol.* 179 (2007) 497–530.
- [20] A.K. Das, C.J. Sung, Y. Zhang, G. Mittal, Ignition delay study of moist hydrogen/oxidizer mixtures using a rapid compression machine, *Int. J. Hydrogen Energy* 37 (2012) 6901–6911.
- [21] B.W. Weber, C.J. Sung, M.W. Renfro, On the uncertainty of temperature estimation in a rapid compression Machine, *Combust. Flame* 162 (2015) 2518–2528.
- [22] D. Lee, S. Hochgreb, Rapid Compression Machines: heat transfer and suppression of corner vortex, *Combust. Flame* 114 (1998) 531–545.
- [23] P. Desgroux, L. Gasnot, L.R. Sochet, Instantaneous temperature measurement in a rapid-compression machine using laser Rayleigh scattering, *Appl. Phys. B Laser Opt.* 61 (1995) 69–72.
- [24] C.J. Sung, H.J. Curran, Using rapid compression machines for chemical kinetics studies, *Prog. Energy Combust. Sci.* 44 (2014) 1–18.
- [25] G. Mittal, C.J. Sung, Aerodynamics inside a rapid compression machine, *Combust. Flame* 145 (2006) 160–180.
- [26] G. Issayev, K. Djebbi, G. Kukkadapu, M. Mehl, S.W. Wagnon, W.J. Pitz, A. Farooq, Experimental and kinetic modeling study of tetralin: a naphtheno-aromatic fuel for gasoline, jet and diesel surrogates, *Proc. Combust. Inst.* (2020) In press, doi:10.1016/j.proci.2020.07.059.
- [27] S. Richter, G. Kukkadapu, C.K. Westbrook, M. Braun-Unkhoff, C. Naumann, U. Riedel, A combined experimental and modeling study of the combustion properties of an isoparaffinic alcohol-to-jet fuel, in preparation.
- [28] J. Badra, A. Farooq, Site-specific reaction rate constant measurements for various secondary and tertiary H-abstraction by OH radicals, *Combust. Flame* 162 (2015) 2034–2044.
- [29] R. Sivaramakrishnan, J.V. Michael, Rate constants for OH with selected large alkanes: shock-tube measurements and an improved group scheme, *J. Phys. Chem. A* 113 (17) (2009) 5047–5060.
- [30] CHEMKIN-PRO 15142 Reaction Design, San Diego, 2015.
- [31] Y. Li, C.W. Zhou, K.P. Somers, K. Zhang, H.J. Curran, The oxidation of 2-butene: a high pressure ignition delay, kinetic modeling study and reactivity comparison with isobutene and 1-butene, *Proc. Combust. Inst.* 36 (1) (2017) 403–411.
- [32] S. Lapointe, R.A. Whitesides, M.J. McNenly, Sparse, iterative simulation methods for one-dimensional laminar flames, *Combust. Flame* 204 (2019) 23–32.
- [33] B. Li, H. Zhang, F.N. Eglolfopoulos, Laminar flame propagation of atmospheric iso-cetane/air and decalin/air mixtures, *Combust. Flame* 161 (2014) 154–161.
- [34] Y. Zhang, K.P. Somers, M. Mehl, W.J. Pitz, R.F. Cracknell, H.J. Curran, Probing the antagonistic effect of toluene as a component in surrogate fuel models at low temperatures and high pressures. A case study of toluene/dimethyl ether mixtures, *Proc. Combust. Inst.* 36 (2017) 413–421.
- [35] Z. Malliotakis, C. Banyon, K. Zhang, S. Wagnon, J.J.R. Henriquez, G. Vourliotakis, C. Keramiotis, M. Founti, F. Mauss, W.J. Pitz, H. Curran, Testing the validity of a mechanism describing the oxidation of binary n-heptane/toluene mixtures at engine operating conditions, *Combust. Flame* 199 (2019) 241–248.
- [36] J. Yanowitz, M.A. Ratcliff, R.L. McCormick, J.D. Taylor, M.J. Murphy, Compendium of experimental cetane numbers, Technical Report NREL/TP-5400-67585, February 2017.
- [37] E.E. Dames, A.S. Rosen, B.W. Weber, C.W. Gao, C.J. Sung, W.H. Green, A detailed combined experimental and theoretical study on dimethyl Ether/Propane blended oxidation, *Combust. Flame* 168 (2016) 310–330.
- [38] M. Mehl, W.J. Pitz, C.K. Westbrook, H.J. Curran, Kinetic modeling of gasoline surrogate components and mixtures under engine conditions, *Proc. Combust. Inst.* 33 (2011) 193–200.
- [39] M. Mehl, H.J. Curran, W.J. Pitz, C.K. Westbrook, Chemical kinetic modeling of component mixtures relevant to gasoline, European Combustion Meeting, 2009.
- [40] J. Bugler, K.P. Somers, E.J. Silke, H.J. Curran, Revisiting the kinetics and thermodynamics of the low-temperature oxidation pathways of alkanes: a case study of the three pentane isomers, *J. Phys. Chem. A* 119 (2015) 7510–7527.